# LOCAL RING DIMENSION DOES NOT INCREASE ALONG LOCALLY QUASI-FINITE MORPHISMS

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*Editorial note.* Counted span is the author's own proof environment, source lines 2987-2999, immediately following the statement at 2979-2985.

*Editorial note.* One cross-chapter pointer (Algebra) and one definitional anchor (Properties of Spaces) are recorded in assembly.external_references.

*Allowed standard prerequisites (editorial).*

- Dimension of the local ring at a point of an algebraic space. Properties of Spaces, Definition spaces-properties-definition-dimension-local-ring, cited at proof lines 5090-5092. Fixes what dim(O_{X,x}) means: via the etale charts.
- Dimension inequality for locally quasi-finite morphisms of schemes. Algebra, Lemma algebra-lemma-dimension-inequality-quasi-finite, cited at proof lines 5107-5108. The imported terminal inequality dim(O_{V,v}) >= dim(O_{U,u}) for the scheme charts.

*Source reference.* Original reference 00QF: focused-seeds/geometry/S9-decent-dimension-local-ring-quasi-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 04NA: focused-seeds/geometry/S9-decent-dimension-local-ring-quasi-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X \to Y$ be a locally quasi-finite morphism of algebraic spaces over $S$. Let $x \in |X|$ with image $y \in |Y|$. Then the dimension of the local ring of $Y$ at $y$ is $\geq$ to the dimension of the local ring of $X$ at $x$.*

*Proof.* The definition of the dimension of the local ring of a point on an algebraic space is given in Properties of Spaces, Definition 04NA. Choose an étale morphism $(V, v) \to (Y, y)$ where $V$ is a scheme. Choose an étale morphism $U \to V \times_Y X$ and a point $u \in U$ mapping to $x \in |X|$ and $v \in V$. Then $U \to V$ is locally quasi-finite and we have to prove that

$$\dim(\mathcal{O}_{V,v}) \geq \dim(\mathcal{O}_{U,u})$$

This is Algebra, Lemma 00QF. □

*Date*: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.